

Narrative

Storytelling
Reception—The Player's Experience of a Story
A Brief History of Literary Theory and Video Games



BioShock (2007) is widely acclaimed for the richness and depth of its narrative

Let us move away from the cultures created by games and their players, and back to the substance of the games themselves. This chapter explores the importance of narrative in video games. Critics, scholars, and parents alike have found many reasons to dislike and criticize video games. One argument is that the medium cannot tell a story, let alone a good story. Much prominent video game theory centers on what many perceive as an incompatibility between gaming and storytelling. This stands in sharp contrast, however, to the avid nine-year-old fan of *Zelda*, who can offer a step-by-step recitation of the adventures of Link as he travels through the land of Hyrule. So we are left with a basic question: are video games stories?

> The rain has been falling like a dense spittle for weeks now, and as you cruise above the metropolis in your spinner, the neon lights blur below like artist's pastels on a sidewalk. Back on the ground to pursue your mission, the babel of cityspeak washes around you. The last warm smile you can remember came from a geisha grinning off the side of a stark high rise, and that was about as genuine as one of those android snakes they're hawking down in Chinatown.[1]

You are a blade runner. As you search for clues on your first mission, life is not kind in the city. The sound of rain accompanies your character's footsteps; you can nearly smell the food as you step into Howie Lee's bar to ask the owner a few questions. Using your mouse, you click on the screen image of Howie and a menu pops up:

- LUCY PHOTO
- RUNCITER CLUES
- EMPLOYEE
- SMALL TALK
- DONE.

These are your options. What will your next step be? You've just returned from the Runciter shop at the other side of L.A., where an animal murder has taken place, and there you have found a pair of chopsticks with the name of the bar you stand in now. Is this just coincidence or an important lead? As you contemplate this, you notice that the bulky employee behind the bar looks a bit like one of the suspects photographed by the shop's security system. You use your mouse to click on "SMALL TALK," and ask Howie Lee about his employee. He cannot tell you anything, so you click on "EMPLOYEE," and the program guides you around to the back to talk to the guy yourself. He claims maybe to have seen the girl whose picture you are showing him, but then he suddenly throws a boiling pot of water toward you and runs away through the back door. Can you dodge the water? Will you follow him or go back to Howie?



Figure 7.1 Blade Runner: Howie Lee's bar and our character

If this were a movie, the main character would probably run after the suspect in the alley; you as the spectator would not give this interchange a second thought. He might even say something witty as he jumped beyond the arc of the boiling water. You would not need to click over and over again on a simplified list of commands. You would sit back and watch the action unfolding in front of you, pondering connections between the story's elements and wondering what might happen next. However, this is a video game, so nothing will happen unless you act. You need to decide whether to move or wait, choose conversation topics, shoot a suspect or not, arrest people or not, and you need always to look for the clues that might be hiding on the edges of the screen. No matter what happens, the outcome is one that you have helped create. You cannot play *Blade Runner* without paying attention to the story, as at any turn you wouldn't know what to do next. *Blade Runner* is *more* than a story—it is a game—but we can't deny that it is *also* a story. Video game genres (and individual video games) are so different from each other that it would be folly to suggest that a story-centered analysis is always a good idea, but there are many cases in which it is, as we will demonstrate in this chapter.

Blade Runner belongs to the genre of adventure games, a direct descendant of text adventures (see Chapter 4), and arguably the genre in which story is most important. There are other subgenres in which stories are very significant, such as role-playing games or so-called action-adventure games. In the beginning of video game history, only text adventures could be said to integrate stories and games (even though many games used the frame of a plot as advertising); but nowadays there is no popular genre that doesn't use some sort of explicit fictional framework. Early video games were simply too abstract for plot to be a central concern—we can only imagine the triangular ship of *Asteroids* being involved in so many plot-heavy adventures.

But as computing processing power grew, the industry tended toward the production of more representationally sophisticated games. Good graphics have translated into painstakingly detailed and sometimes beautiful worlds, while better AIs allow for more advanced responses to players' actions, including more interesting nonplayer characters, better simulations, and the possibility of handling thousands of players at the same time in huge multiplayer worlds—all of these advances make the game world seem more alive, and each, in turn, changes the possibilities for storytelling.

Narrative is so pervasive that there are video games based on nothing more than a tangential connection to stories in other media; the most popular of these are games directly created from other cultural products, such as films. These games can sometimes have a narrative element, but often the film ends up being a thematic excuse for a platform or action game. Consider, for example, *Star Wars Episode I: Racer*, developed in connection with the movie *The Phantom Menace*. The game draws on the fictional universe of the *Star Wars* series but is based on only one scene from *The Phantom Menace*. As a typical racer game, it contains almost no story, but it nevertheless prompts the player to "take the controls as Jedi-to-be Anakin Skywalker."[2]

There is only a suggestion that the player might frame her actions within the game as part of this specific fictional universe; the act of playing itself will not reproduce the sequence of events as in the movie but allows for many other combinations.

Even in genres where stories are not part of the gameplay at all, such as strategy or simulation, fictional worlds prompt players to imagine that their actions take place within a meaningful narrative frame. For example, Microsoft's successful *Age of Empires* is a strategy game based on the management of resources in the classic setting of "civilization-building," in which your tribe has to develop from nothing to an empire, and compete with others for world domination. The theme of advancing your empire gives a reason for abstract resource management and provides the frame for the game's graphics, settings, and action. Themes and plots—however vague—enable players to figure out game interfaces and the rules of the game: what actions are available to me as a player? And that is why most games use them. *Age of Empires* has a final goal: *advance your culture and dominate all others*, but even video games without any goals, such as the toy-like *The Sims*, coat the simulation with a fictional varnish. In the promotional materials for the game, you are invited to play God in an adult version of a doll's house.[3]

> Create simulated people and build their homes, then help your Sims pursue careers, earn money, make friends, and find romance—or totally mess up their lives. Test your "people skills" as you deal with family, friends, careers, and chaos! There is no right or wrong way to play the game.[4]

Clearly, stories are everywhere. And we will use the general term "narrative video games" to refer to any games in which stories play a significant role. But to avoid the confusion that has haunted many discussions about the role of narrative in games, let us define our concepts.[5] So far, we have been using the words "story," "narrative," "fictional world," and "fiction" in what seems like an interchangeable manner, something that would no doubt horrify many literary scholars.

"Narrative" can be defined as a succession of events. Its basic components are the chronological order of the events themselves (story), their verbal or visual representation (text), and the act of telling or writing (narration).[6] For example, the novel *Dracula* tells us a story of how an ancient vampire comes to England, wreaks havoc in a small community, and is finally destroyed. This story—which is not always clearly or chronologically told—is found in the text, written by Bram Stoker. Bram Stoker (whom we cannot see in the text) creates the narration, both through his own descriptions and through the voices of the characters, who "speak" through their letters and other writings.

Technically "story" and "plot" are "ingredients" of narrative, understood as a succession of events. However, in this chapter, "story" is more of a loose term, referring to fictional worlds projected by the games, unless it is described as "linear story." "Plots" or "narratives" are a scripted succession

of events that the player has to perform in a specific order. (In *Blade Runner* you have to find the chopsticks before you can go to Howie Lee's bar, for example.) The events that make up the whole narrative or plot of this kind of game usually allow minimal flexibility in the order of completion.

The other components of narrative—text and narration—are not as essential to our discussion of video games, mainly because these games lack a static text. No matter how many times a book is read, by no matter how many different people, the text is always the same; but in a video game, no two game sessions will be exactly the same. As for narration, the nonfixed text of video games makes it difficult for them to have a single telling; the exception proves the rule here, as there is a very short list of games, such as *Prince of Persia: Sands of Time* or *Max Payne*, that feature a voice-over narration of the type you might have in a movie, which frames the player's actions within a narrative framework that has already been fixed in advance. But this is a trick, of course, because the voice-over only concerns itself with the main plotlines of the story, and it is still possible to have different playing sessions in which minor things are done slightly differently.

As we can see, these terms are concerned with the structural description of a narrative. We should also mention the broader terms "fiction" and "fictional worlds." "Fiction" is not an uncontested term, but we can make a pragmatic distinction between "fiction" as events that have not occurred in "real life" (the basis for novels, films), and "nonfiction" as documentation of events that have occurred (as seen in news items, autobiographies, and film documentaries). Nearly all video game stories are fictional, although there are also games, such as *Battlefield 1942*, based on real World War II battles, which simulate historical events.

A fictional world is an imaginary construct created by the descriptions of a text. Readers—and viewers, and players—infer that there is a make-believe universe in which the events they are reading about (or viewing or performing) make sense. For example, from reading the tale of *Little Red Riding Hood* we can imagine a pre-modern world where people lived close to forests, had tight family relations, and where animals could talk. From our own experience, we know that grandmothers and granddaughters are usually fond of each other, and that little girls shouldn't talk to strangers. This is much less detailed than, say, the fictional world that J.R.R. Tolkien creates in *The Lord of the Rings*, where he invents languages and carefully describes continents, races, animals, plants, and an entire mythology.

However, these two narratives can affect us similarly as readers; the point is that we do not need much descriptive prompting in order to ascertain which kinds of events make sense in a concrete fictional world. As we read, or view, or play, we unconsciously and immediately apply what we know from our own world if it is a realistic setting, and from other fictional worlds if it is not. For example, the player of *Half-Life* assumes the role of a scientist trapped in a secret research facility, where an experiment has gone horribly wrong and brought evil aliens into our world. The player quickly recognizes the science fiction-conspiracy theme (from novels or Hollywood movies)

and can almost immediately guess the plot to come: the government will try to eliminate him and all other surviving witnesses. *Half-Life* is full of cues of all kinds: visual, in the dialogues, and in the kinds of actions that we can (and can't) perform with our character. The fictional world is immediately recognizable, and thus believable, because it is similar to many stories we have seen and read before.

Most games have a fictional world, however minimal, and the success of this imaginary creation nearly always influences the player's enjoyment of the game. *Star Wars Episode I: Racer*, as we saw earlier, wouldn't be as engaging if instead of pods and the characters from the *Phantom Menace* movie it featured cars and an unknown set of colored animals as characters. With the visual clues from the movie, the player can imagine her actions as loosely connected to a broader *Star Wars* story outside the game.

Narratives are made of events, and usually contain settings and characters, but both of these ingredients can appear on their own, without being tied to a specific narrative, so that players can imagine how setting or characters fit into the fictional world, relate them to an external story (like *Star Wars*), or simply use them in order narratively to thematize their enjoyment of the game (in *Super Monkey Ball*, for example).

We should point out that it is easy to confuse fictional worlds with narratives. This is because one of the ways we understand a narrative is by filling in the gaps: we postulate connections between events, we interpret the motives of characters, and so on. In other words, we project an imaginary world. In the same confusing way, when video game designers talk about narrative, they usually refer to the introduction of elements that prompt the player into imagining fictional worlds—which could be anything from excellent characters to detailed environments. Many designers abhor the scripting of programmed sequences of events in games, which would actually form a narrative in the literary sense.

Much of the scholarly discussion around narrative and video games deals with the perceived difficulty of combining a playing experience that feels free with the necessary constraints of a narrative structure. In other words, the problem of letting players act freely while ensuring that their actions produce an interesting story.

This chapter will introduce some of the concepts that are important when looking at video games from a narrative perspective—as regards both narrative as a sequence of events and the creation of fictional worlds—in the section called "Storytelling." This section also argues that story and gameplay are integrated in the playing experience, as we will see from the analysis of a particular game. The second part of the chapter, "A Brief History of Literary Theory and Video Games," provides an overview of major theoretical issues in discussion in the video game narrative community. This chapter should give you a fairly extensive idea of the work that has been done in the area, as well as hopefully demonstrating that a consideration of narrative is useful for understanding video games.

STORYTELLING

We will here introduce interesting elements of narrative that are relevant to video games. We have divided the concept of storytelling into three broad categories, which can be thought of as the "who" and the "what," and the "how/why" of a narrative. The categories are "the fictional world," which includes the story's settings and actors; the "mechanics" of the narrative, or how the action of the story is organized; and the **reception**, or the way in which players experience the story and figure out the causality.

Reception (studies): In literature, the study of a reader's understanding of the text. In video games, the study of the player's experience of playing a game.

The Fictional World: Settings and Actors

Most video games are given a setting that not only helps the player frame her actions but is also engaging enough that it has some value in itself. Players of *Myst* marveled at the beautiful landscapes and eerie atmosphere of the deserted islands. Fans of *The Last Express* no doubt still remember the animated life of the train where the game takes place, full of characters that go on about their business without regard for the protagonist. And those players steeped in *Counter-Strike*, where enemies can appear in any doorway, and where danger lurks around every corner, may even find themselves fantasizing about combat tactics as they walk down corridors in real life. The list of fictional worlds is endless—from the mysterious building where the old *Donkey Kong* takes place (full of jumps and stairs and barrels and dangers for our little Mario) to the terrifying village of *Silent Hill*—and video games have relied upon these imagined spaces from their inception.

But what exactly is the relationship between fictional game worlds and the game experience? Lisbeth Klastrup describes the differences between game worlds and social worlds, and argues that a game world's objects and space are all organized around the act of play.[7] Many of us have tried to open doors and windows that would not move in *Blade Runner*, or tried to climb the trees of *EverQuest*; the fictional world of games is like the stage of a play—it is meant for action, but many of the elements are there to be seen and not used. Only those objects directly related to gameplay will be "usable," although quite a few contemporary games with advanced physics offer the possibility of interacting with objects unrelated to the story. Games indicate which objects are relevant in different ways. In *Blade Runner*, for example, the cursor changes color when the player moves it over a usable object. Others simply try to make as many objects "alive" as possible, in simulations that provide the illusion of a virtual reality. In *Grand Theft Auto V*, we move around in a city where all cars can be stolen, all people beaten up, and a lot of other things done that strictly speaking have nothing to do with the missions we are given. The city, it seems, lives even without us. Although this is, of course, an illusion, as we can only interact with the different kinds of objects in one predetermined way: we can beat people up but not kiss them, for example. Designers have divided opinions as to what kind of world

works best, but players' expectations are likely molded by a game's genre: we are more likely to try and explore every nook and cranny of an adventure game as we look for clues, and less likely to care about the surroundings in an action game.

The most important component of a game world is the gamespace,[8] understood as the setting for the gameplay. Gamespaces are not realistic but reductive: they reproduce some features of the real world but create their own rules in order to facilitate gameplay (and to reduce the processing power required by a computer to run the game). For example, in *Myst* (see Chapter 4), the world around us seems open, but in actuality we are forced to follow a very specific route through the landscape, and can't stray off the path or explore.

Espen Aarseth has looked at different kinds of spatial representation in video game history: from two-dimensional to three-dimensional, from open landscapes to landscapes where you can only move in a single direction, and from indoor corridors to space travel and everything in between. He argues that spatial representation is the most innovative aspect of video games. Following on from his main point—that the spaces of video games are fundamentally different from real spaces—he makes an interesting distinction between space in single- and multiplayer games. Single-player games, as in the case of *Myst*, for example, often have landscapes that seem to be open but are actually very restrictive. This is necessary because the game orchestrates the movement of the player in one direction, toward a final goal. Multiplayer landscapes, however, have to be open so that all players can move freely, and so that no player starts with an advantage. The challenge in these games is not the landscape (as it would be in a platform game) but the other players.[9]

Gordon Calleja has examined the importance of gamespace as "a powerful factor in engaging players and giving them the sense that they are inhabiting a place rather than merely perceiving a representation of space."[10]

One of the most contested issues in relation to the fictional world of games has been cut-scenes, cinematic sequences used to relay information to the player. Many games with a sophisticated story use this technique to situate player actions in a fictional world that can thus be described with great authorial control. Play is interrupted and we watch a "film" where the game characters interact or something happens that is out of our control. Cut-scenes are used by designers to create narrative in a variety of ways.

First, they introduce a central narrative tension. Many arcade games used animated cut-scenes to introduce the central conflict of the game. These scenes were brief and served to give the player a chance to get ready for the actual gameplay. As the need to get the player through the game quickly fell away, introductory cut-scenes grew longer, and today's games often have extended movie sequences that introduce characters, and set the scene and the mood.



Figure 7.2 New Zealand Story (1989) cut-scene, where the evil walrus kidnaps the kiwi family but one bird escapes

Second, cut-scenes shape the narrative in a certain direction. They function to ensure that the game protagonist makes certain choices. The player is stripped of his or her influence and the narrative is moved along. In the game *Gabriel Knight III*, for instance, Gabriel walks through a door, triggering a cut-scene in which the player has no control over Gabriel's actions.

Third, cut-scenes compensate for missing game narrative. Many games mark the passing of time within the game using cut-scenes. Sometimes these scenes illustrate a journey; for instance, when the player of *Prince of Persia: Sands of Time* has performed a certain series of tasks the game jumps to a longish cut-scene showing the consequences of the prince having procured the Dagger of Time.

Fourth, they associate the game with contemporary cinema aesthetics as in *Max Payne II*, for example to build a richer game experience through an emotional tie.

Fifth, cut-scenes provide the player with information. Some modern games have integrated cut-scenes much more closely with the gameplay than was previously the norm. Here the cinematics are used to convey useful information, often to serve as a kind of establishing shot describing the layout of a location. As the protagonist enters a new area, the "camera" may swoop through the scene showing the placement of enemies and objects.



Figure 7.3 Max Payne II (2003) cut-scene, where the player is emotionally tied to the main character

The rationale for using cut-scenes in games is disputed. Some hold it to be a case of cinema envy; some argue the technique compensates for game design incompetence, so that the fictional worlds are shown passively instead of letting the player discover them through gameplay.[11]

Others, however, see the attack on cut-scenes as a radical or purist attempt to (over)emphasize what is special about games.[12] In his paper "In Defense of Cutscenes," Rune Klevjer argues that cut-scenes are a manifestation of the author's voice in the creation of the diegetic world, understood as "a fictional world, created by discourse."[13] The importance of cut-scenes is not only retrospective; as we have seen, it also frames the world in which play takes place:

> The cut-scene may indeed be a narrative of re-telling but more importantly: it is a narrative of pre-telling, paving the way for the mimetic event, making it a part of a narrative act, which does not take place after, but before the event. The cut-scene casts its meanings forward, strengthening the diegetic, rhetorical dimensions of the event to come.[14]

Apart from the settings manifested in the gamespaces and conveyed by cut-scenes, the other important ingredient when building fictional worlds is the characters that will populate them. We have talked about "actors" before, because many of these characters are capable of action, and indeed of interacting with the player. If we take a dictionary definition, characters are "the people that the film, book, or play is *about*."[15]

It seems that games (when they have characters) would participate in both modes—the narrative and the dramatic: sometimes we read and imagine worlds (as in adventure games), sometimes we view worlds unfolding (as in cut-scenes in adventure games), but the revolution offered by video games is that we can play some characters ourselves, and that other characters will react to us and do things that we can respond to. Characters in games are not just the people that the game is *about* but also the people who are making action happen and thus producing different stories. Mark J. P. Wolf considers characters to be the driving force of narrative development in video games. He proposes that they be categorized by their function: playing characters, computer-controlled characters and computer-controlled characters with no action in the game (helpers, hinderers, beneficiaries, neutral characters, or narrators).[16]

We propose our own typology of characters in video games, organized according to the extent to which players can interact with them:

- *Stage characters*: Just part of the scenario, moving around but with no personality or function in the game. We cannot interact with them in any way.
- *Functional characters*: Like stage characters but with a general function in the game; for example, in *Grand Theft Auto III* we can attack or be attacked by people walking past, and in *Vampire: The Masquerade—Redemption*, our vampire character once in a while needs to drink blood from anonymous mortals.
- *Cast characters*: Characters with a particular function in the game related to the story. They have personality to different degrees: a minimal amount in the case of the *Blade Runner* bartender we have to interrogate, for example, and a large amount in the case of our friend—later enemy—Sephiroth in *Final Fantasy VII*. Such characters have their own agendas, and can be considered in Proppian terms,[17] as we will explain.
- *Player characters*: The character controlled by the player (except in cut-scenes); we can usually control his actions but his motivations and missions are decided by the story.

Even though all characters can be memorable in their own way, the player character is certainly the crucial one for the player experience. Designer Toby Gard has proposed a typology of player characters, with three categories according to how easy it is for players to identify with them: from avatars to actors, with role-playing game characters occupying a middle position:

> *Avatars* are a non-intrusive representation of ourselves, *actors* are always part of a story (or have a story, albeit minimal sometimes), and *roleplaying* characters have very different abilities that we can raise according to our performance.[18]

Typically, an avatar has no name and cannot be seen, as the game view is first person, so that the player merges with the character, as in *Unreal*, for example. Actors can usually be seen in third-person view and have their own biography integrated with the game story, for instance, Lara Croft of *Tomb Raider*. Role-playing characters are "made" by players, who choose their name, appearance, and abilities, and can often also choose between the first- and third-person views. This last category should in theory offer more room for identification as the player constructs the identity of her own character in play. However, in practice it is hard to really role-play and develop a character within the context of MMORPGs, as MacCallum-Stewart and Parsler have demonstrated.[19]

Tosca has added a fourth category: "iconic" characters. These are characters, such as the *Super Monkey Ball* monkeys, who are empty of personality but allow for the collective channeling of stereotyped expressions in the game performance, such as happiness in victory or dismay when losing.[20]

The importance of being able to identify with a game's characters is something of a recurrent theme in video game design manuals; there is a belief that the stronger the personality of the character, the easier it is for a player to feel alienated from it.[21] The implication—that designers should create characters with only vague attributes—is the opposite of a typical modern literary perspective, where characters are praised for their vivid uniqueness. The fear of alienation is also contradicted by some very popular video game characters, such as Lara Croft, who have very distinctive personalities.

The last point we will consider in relation to characters is that of the creation of better (meaning more active) nonplayer characters, a problem directly related to the development of artificial intelligence[22] and one that has worried many developers for a long time. Brenda Laurel is in fact optimistic, and proposes simplification rather than more complexity:

> the fact is that, thanks to well-internalized dramatic convention, we can enjoy (and believe in) even one-dimensional dramatic characters. In fact, when a minor dramatic character possesses only one or two actionable traits, audience members will impute elaborate histories and motivations as needed to make it believable.[23]

Laurel builds on the work of Meehan,[24] Lebowitz,[25] and others who have established guidelines to create "functional and entertaining characters from a small cluster of well-conceived traits that are realized as goal-formulating and problem-solving styles."[26] Despite constant technological advancements and more and more complex games, good characters do not necessarily have to be lifelike. The techniques mentioned above can allow for the creation of a very interesting array of characters of all categories (stage, functional, cast, and player characters). And of course there is the fascinating possibility of interacting with the characters of fellow gamers in

multiplayer titles, an activity that has not been thoroughly explored in the research.

Mechanics: Organizing Narrative Action

We now turn to an examination of what constitutes narrative action in video games, and ways of implementing it. A central preoccupation of game designers is the problem of linearity—how to get a player to move through a game in one particular way, at the same time ensuring that it is interesting. As the narrative possibilities of video games have become more complex, so too has the debate over the mechanics of the narrative. Designers now generally accept that forcing the player through the fictional world does not make a good video game, no matter how compelling the narrative. But if you don't force your player to do things, then how do you create a plot? This is the central question for narrative mechanics.

The basic concept designers use to organize narrative action is "branching." We can define branching as the existence of multiple paths in a narration. Branching leads to the problem of managing the exponential growth of nodes (or individual text spaces) in a narrative. This has been one of the most debated issues among hypertext theorists since the early 1990s. Faced with the same problems, video game designers have generally allowed for moderate branching while implementing plot bottlenecks, through which all players have to pass in order for the story to advance. For example, in *Gabriel Knight: Sins of the Father*, the player has to visit a certain number of locations in order to finish each day; she can do it in different order, but in the end, the day will finish in the same way. Jonas Heide Smith has illustrated the differences between the standard narrative progression of linear fiction (including novels and films) and that of interactive fiction and narrative games.

As illustrated in Figure 7.4, traditional fiction goes only one way—toward a resolution—and does not allow for interruptions or cuts. A traditional narration works because it is a continuous line that plays with a reader's (or viewer's) expectations and orchestrates their emotional trip from beginning to end, controlling important points such as the point of no return and the climax. If we applied the same model to a video game, we could not allow the player to do anything, as any deviation or delay would ruin our carefully planned emotional curve. Imagine the player dying in the middle of the narrative! But a game with such a narrative model would

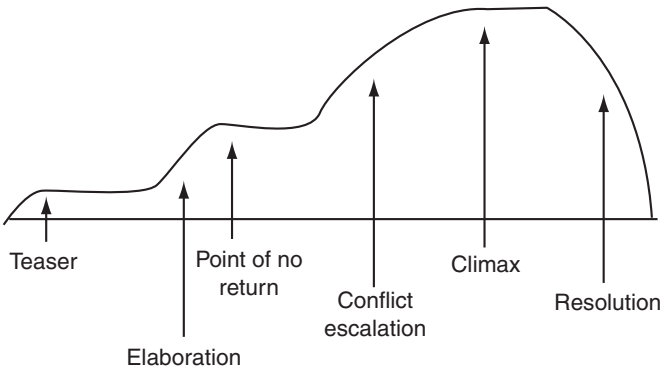


Figure 7.4 A model of classical linear fiction that shows how the story is built up in stages

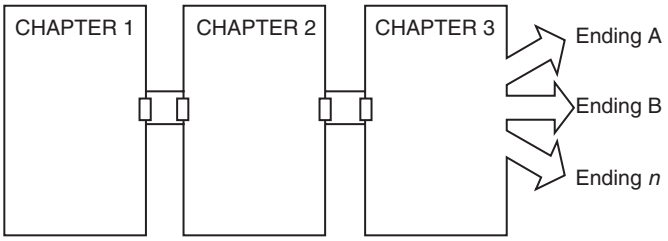


Figure 7.5 A model of interactive fiction that shows the branching of the story

not be much of a game, which is why designers opt for another model (see Figure 7.5).

In this kind of fiction—typical of adventure and action-adventure games—there is no continuous curve. The player has to solve puzzles (or find objects, win battles, or talk to nonplayer characters) within each chapter; usually there is some flexibility as to the order of the tasks, offering a small sense of freedom. In the absence of an emotional curve, designers rely on the emotional satisfaction—the "unambiguous sense of victory"—that solving the puzzles gives the player.[27,28] But ultimately the player is *solving* a story instead of actively *creating* it. The successive chapters work in a cumulative way so that at the end of each a climax or resolution is provided, usually preceded by a difficult "boss fight" to give a greater sense of achievement. As mentioned in Chapter 5 on video game aesthetics, Jesper Juul calls these "progression" games, because the player has to complete a certain sequence of actions in order to get to the end—or endings, as sometimes these games allow for different resolutions according to the player's actions.

Aside from progression games, the other dominant narrative structure—as defined by Juul—is "emergence" games. This type of game follows nothing like the classical linear structure; instead of a prestructured sequence, there is a more active artificial intelligence (AI), in which each object has behaviors. For example, in a progression game, the dragon will always attack the player when she steps into the cave, but in an emergence game this might depend on how the player behaves toward the dragon, which is a more "active" object with a few possible different responses. For both Juul and Smith, an emergent structure is preferable, as it gives players a sense of freedom. Smith calls it "deistic narration," a kind of object-oriented narration in which game designers become architects of space and situations:

> An example: A virtual living room is designed, the avatar is placed on a couch, and a dragon is placed under the couch. What we then *have* is not a story but a story is what we may *get*. What we have is a starting position with narrative potential but without direction.[29]

The key to successful mechanics is to make players feel that they are contributing to creating a plot; the most successful narrative experiences happen in games where our actions have noticeable plot consequences.

Beneath the narrative structure of an entire game, designers must also manage the smaller sequences of events that make up the arc of a game's plot. To explore these building blocks of narrative, some game theorists

have investigated **quests**.[30] Quests are small "missions" that players must perform; they structure a game's action and create opportunities for storytelling.

From the designer's point of view, a quest is a set of parameters in the game world (making use of the game's rules and gameplay) that creates a challenge for the player. From the player's point of view, a quest is a set of specific instructions for action; these can be a general goal (overthrow the evil king) or extremely precise (take this bucket to the well, fill it up, and bring it back to me). After the quest has been completed, it can be narrated as a story. Quests are a way of structuring the events that constitute a game, as they manifest causality at two levels. On a semantic level, quests demonstrate how and why a player's actions are connected to each other and to the end of the game's story; on a structural level, quests embody the cause-and-effect relationship between a plan of action and its results, or between the interaction of objects and events. These two levels can be perceived by both the player and the designer, and if the quests are well built, they enhance a player's emotional engagement. Ideally, quests are the glue that holds world, rules, and themes together in a meaningful way.

Most games with a narrative component employ quests—not only adventure games but also role-playing games or multiplayer role-playing games (where the massive presence of player characters does not leave a lot of room for designer-controlled story production). In *World of Warcraft*, for example, at any given time of day or night, thousands of players around the globe are running around their fictional worlds, engrossed in their particular quests: find a book, kill a monster, take a letter to another city. In adventure games, quests are often defined in terms of puzzles or enigmas that have to be solved by collecting objects, using information, and rearranging machinery.

Game designers should ideally avoid quests that are meaningless for the player. The two levels on which quests operate (the semantic and the structural, as discussed) are also the two levels on which quests can fail. Quests generally do not work because they fail to integrate the storytelling elements in a particular game: if quests feel disconnected from the plot, the game world, or our characters, chances are that the bridging of the semantic and structural levels has not succeeded. There seems to be a fairly fixed and small number of typical quests that many games repeat (the exchange, the breach of contract, the discovery of the traitor, and saving the kingdom are among the most popular), and we might ask ourselves if it is desirable to explore new lands and devise new versions of these quests or search for entirely different ones. As Jill Walker suggests in her analysis of *World of Warcraft* quests,[31] some of the classic quests, such as taking a letter from one character to another, can feel more significant if they reveal relationships between the nonplaying characters that populate the world of the game. In other words, designers need not invest a lot of effort in making player characters a part of the quest stories—a narrative can also be richer if the static world is better fleshed out than in most current games. This is also

Jeff Howard's point in his *Quests: Design, Theory and History in Games and Narratives*.[32] More recently, Tanya Krzywinska has looked at the episodic narrative elements of quest games through the repetition of certain patterns ("beats"), using *World of Warcraft* as an example. Here, quests are seen as micronarratives set within an overarching multithreaded narration.[33] These authors have taken the analysis of quests to a level of greater sophistication.

Marie-Laure Ryan[34] has over the years extensively worked with narrative structures, and has mapped out nine fundamental narrative structures that are described below:

1. *The complete graph structure*: In this narrative structure, all the paths are bidirectional and all nodes are linked to each other. The reader has total freedom to navigate. It is almost impossible to guarantee a coherent narrative as the reader has full control of what to do.
2. *The network structure*: In this narrative structure, paths can be either unidirectional or bidirectional. This is the standard structure of literary hypertexts. The reader's navigation is neither entirely free nor limited to one course.
3. *The tree structure*: In this narrative structure, all paths are unidirectional, and progression produces a plot. The reader's progression is carefully controlled, so any choice results in a well-formed story. This structure is sometimes used in adventure games, although the maze structure explained later is more common.
4. *The vector with side branches*: In this narrative structure, there is one main story that progresses chronologically. However, the reader is allowed or even encouraged to make small detours, which in video games are often referred to as "side quests." The structure resembles that of sightseeing with a guide.
5. *The maze structure*: This narrative is the one mostly found in adventure games, where the players have to find the right path to successfully complete the game. There are more ways to reach the end, but you will always get to the same end. The story will be quite coherent as the game's design sets down blockers between each chapter, ensuring that the players go through the key nodes to progress the story appropriately.
6. *The directed network*: This narrative structure is more chaotic, but strong attempts are made constantly to maintain some coherence as the player progresses. The player has some freedom to connect the experiences of the journey in different ways, forming a somewhat unique narrative. The player feels more empowered as the decisions made during the story influence what happens in the later narrative.
7. *The hidden story structure*: This narrative structure is more complex, as it operates with a fixed narrative that the players slowly uncover as they progress through the game. The game is a reconstruction of a fixed narrative that develops into a new narrative. This worked very well in the first *Max Payne* game, where you slowly uncover what happened to the protagonist's family.

8. *The braided plot*: This narrative structure allows the player to experience different perspectives on an unfolding narrative by changing the protagonist and location. This can be by playing characters with different abilities or access to special knowledge.
9. *Action space, epic wandering, and story-world*: In this narrative structure the player can interact on the macro level to choose different incidents (micro-level). Once the player is on the micro-level the narrative is locked down, and a narrative plays out with limited agency for the player.

Not many progression games manage to give players a sense of freedom, but there are some notable attempts, such as *Blade Runner*, mentioned previously. In this game, there are almost no puzzles; instead the player goes through different scenarios in search of clues. The player has a variety of options each time he confronts a replicant (the game's initial villains). Arresting, killing, or letting them go free prompts complex responses; ultimately, the player's actions decide if she turns out to be a replicant herself—one of the most interesting questions raised by the film the game is based on.

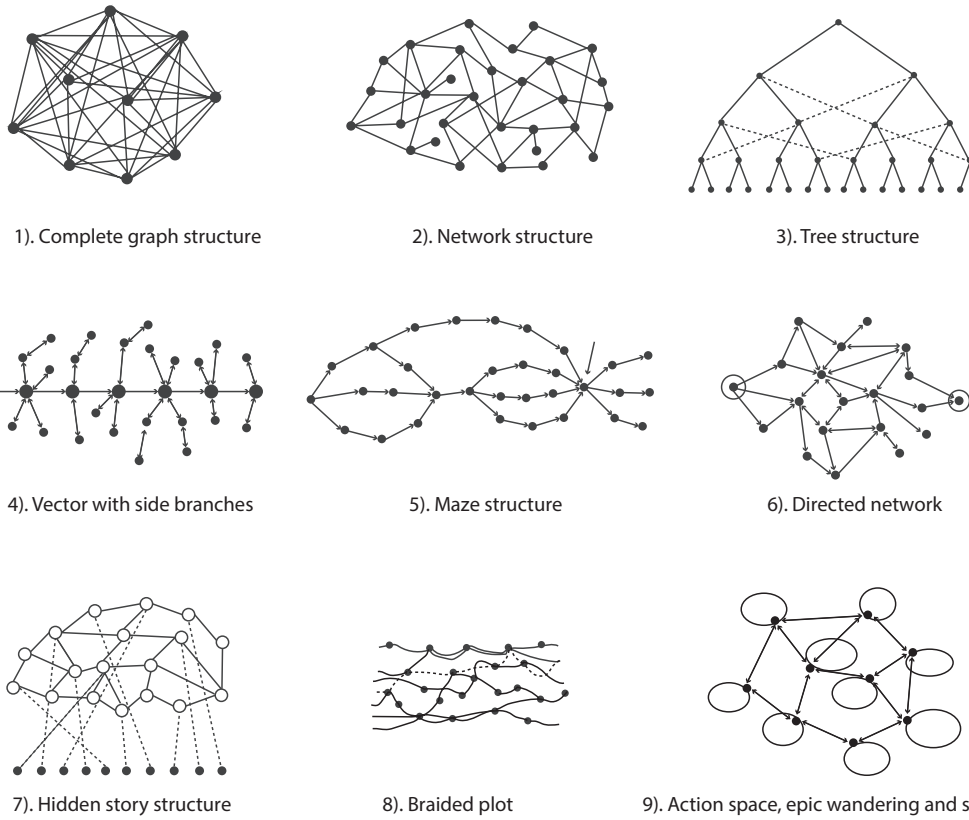


Figure 7.6 A variety of different interactive narratives and how they can fold into each other. (Source: Ryan, 2001b, pp. 247–255, figures 4–11)

Reception—The Player's Experience of a Story

"Reader response criticism" or "reception theory" is a branch of literary theory that focuses on the experience of readers as they interact with texts, and tries to articulate the nature of the reading activity. It is a useful perspective from which to study players' experiences with narrative video games. In what follows, we will argue that a reception theory-based analysis can explain the way that narrative and gameplay together determine the player experience in games that make use of stories. This approach can also answer the question: do we need to pay attention to the story in order to play these kinds of games? A close analysis of the game *Resident Evil Code: Veronica X*[35] will serve to introduce these ideas.

Resident Evil Code: Veronica X is the fourth major title in the very popular *Resident Evil* series. It is somewhere between the adventure genre and the "survival horror" subgenre. Along with other titles like *Silent Hill* and *Alone in the Dark* that fit this hybrid category, the player controls a character who has to get out of some enclosed place, solving puzzles and destroying horrific monsters along the way.

The story in *Resident Evil Code: Veronica X* is about a young woman, Claire Redfield, who travels to Europe to find her brother, Chris. Both siblings are survivors of previous games, and enemies of the evil "Umbrella Corporation," which experiments with biochemical viruses that turn people into zombies and monsters. At the game's outset, Claire is captured by the company and sent to a prison on a monster-infested island, from which she has to escape. Throughout the game, the player alternates between controlling three characters and uncovers yet more of the evil corporation's dark secrets.

Let's begin our analysis with an initial concept, the "literary repertoire," because it will help us understand how readers (in this case, players) can start interpreting a story successfully—that is, how they get into it. The literary repertoire is described by Wolfgang Iser as "the familiar territory within the text,"[36] which can include anything that the reader might already know—references to earlier works, social norms, and historical events. This repertoire brings context into the discussion without opening the door to excessively subjective interpretations or psychological particularities: the reader can only actualize[37] what is already in the text. Understanding the repertoire is a matter of competence, and it can affect both content (for example, recognizing a quote) or form (being able to interpret the conventions of the horror genre). In the case of *Resident Evil Code: Veronica X*, the repertoire includes (among other things):

- *Knowledge of the survival horror genre of video games*: This includes basic premises such as kill all monsters; pick up all objects to solve puzzles; there is a boss, or monster, that is especially difficult to kill at the end of each level or area.

- *Knowledge of the horror B-movies that inspired this kind of game*: These movies place the game in the right cultural context but also offer useful tips; horror devotees all know, for example, that zombies move really slowly, so you can probably dodge them instead of always fighting them (something that will prove particularly useful, since it is easy to run out of ammunition).
- *Knowledge of how to cheat since the player can become stuck, which stops the game*: Savvy players will know to consult a "walkthrough," the detailed turn-by-turn summaries of games discussed earlier. They will also know where to find "cheats," such as pirate programs to optimize their performance in the game. (Note that just as in the case of B-movies, here we are stretching the repertoire to include meta-elements—or elements beyond the game itself—but such "cheats" are now an established part of game communities.)

The repertoire is activated by "clues" in the game (everything from the creepy sounds when you go up the stairs to the dark rain on the fresh tombs) and outside the game (such as the game cover, which shows a zombie reflection in the eyes of the protagonist) which indicate that we are immersed in a survival horror game. Even if we were to start the game knowing nothing about it, we would still be able to summon the right repertoire—the right mental category—soon enough, as we immediately have to face the creaking stairs, and a few minutes later we find ourselves in the first graveyard surrounded by zombies.

If players fail to recall the right repertoire—imagine, perhaps, that we had never played a similar game, or never seen a horror movie—the game cannot be appropriately enjoyed. In the first scene, for example, an injured soldier helps Claire out of her cell. If we wrongly summon the repertoire of "love story" to try to explain this, we will concentrate on the fact that he is wounded, and waste all our time either staying in the cell with him, or trying to find medicine to cure him (which will not happen until we have done many other things in the game).

In reader response criticism, the act of reading is vital, as it transforms the literary text from artifact into aesthetic object. In other words, a text begins as a "dead" object with a lot of potential meanings that only become actualized through reading. Wolfgang Iser attempts to explain how this happens through his theory of "filling in the gaps," in which a reader starts from what the text says and figures out what it does not say. For example, if the text mentions that two new characters who meet in an elevator exchange hateful glances, the reader will wonder what the reason behind it is: maybe they know each other already and have had a fight; maybe they are ex-lovers; maybe they are fans of opposing football teams. As readers, we tend to try and confirm these hypotheses as we read.[38]

While playing a video game, our mind is busy with the story level and the action level at the same time. The first of these, which we experience on

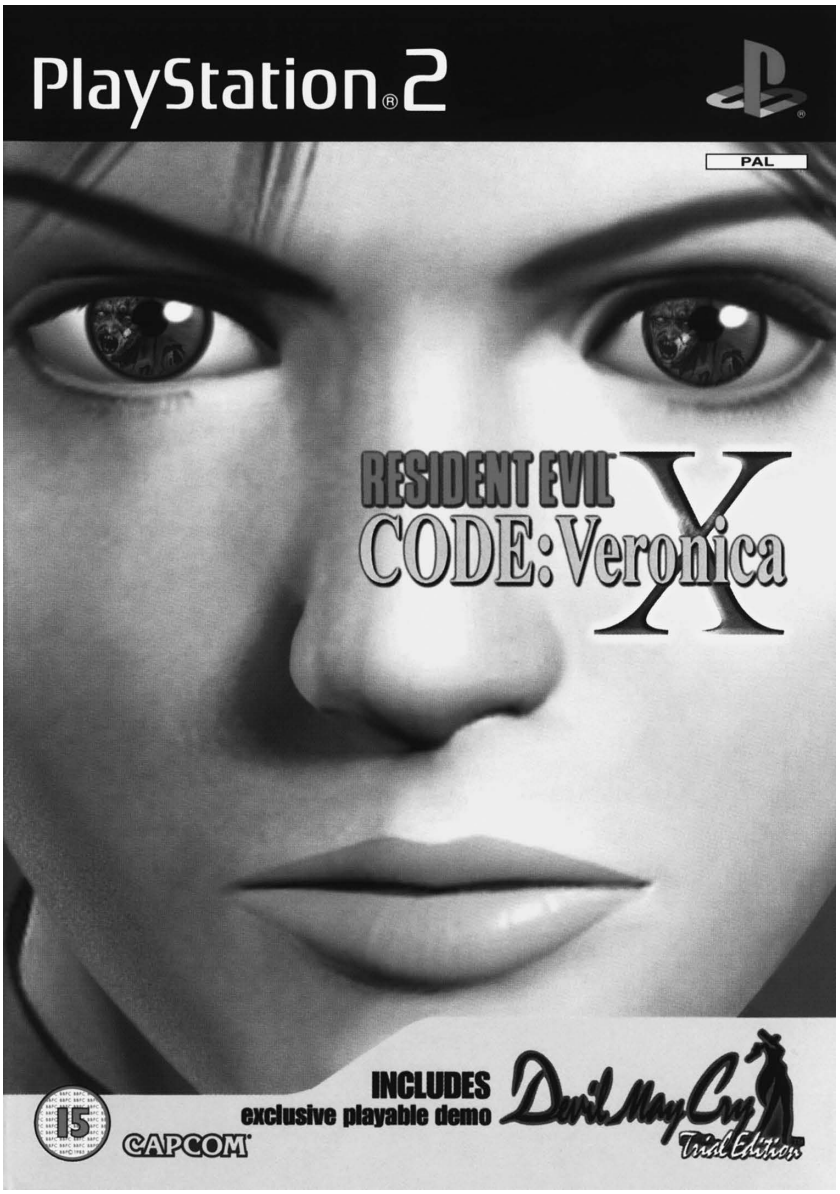


Figure 7.7 The cover of the European version of *Resident Evil Code: Veronica X* for PlayStation

the fly, can be narrated afterwards (it is "tellable"[39]) and makes sense as a story (complete with character motivation and feelings); the second is about solving action problems, and if it was to be narrated it would be as a "walk-through." To illustrate this, let us enter into *Code: Veronica*. The game opens with a cut-scene in which our character, Claire, is captured by the Umbrella Corporation men. The following description of this scene is written by Dan Birlew and Thomas Wilde, who have transposed the *Resident Evil* games into a series of stories.

Story Level

> Slowly, someone clutching his stomach shambles into the room and stands outside Claire's cell door. Claire uses her lighter to see who it is, and is surprised to see the face of the man who took her prisoner in Paris.
>
> The man unlocks her cell and opens the door. As Claire hesitantly steps outside her cell, he slumps into a nearby chair and pulls an empty bottle of medicine out of his pocket. He throws it against the floor in frustration. Not looking up, he tells Claire that the place is finished. They've been attacked by what he thinks is a "special forces team." Claire's free to leave the prison grounds, but he warns her that she has no chance of getting off the island.[40]

The gaps here open and close as they do in any narrative: Who is the guy? (open)—Umbrella guard who captured me (close); Why is he opening the door? Maybe he will kill me, maybe he is a spy (open)—he opens it because the base is under attack and wants to give me a chance (close); Do I believe him? (open) Do I have a chance? (open) Who is attacking us? (open) We could continue, but you get the idea: just as when reading a story, the gamer actively works to fill in the gaps.

However, unlike in a novel (or a movie), as the story grows and changes in our minds we also have to act. This walkthrough demonstrates the choices a player must make as they decipher the game's clues.

Action Level

> CELLBLOCK ROOM—We are in a cellblock. Start off by going to your item screen. Go to the LIGHTER you see and use "Select" with the action button. This will prompt a cut-scene. After the cut-scene, go back into the cellblock and grab the ever important GREEN HERB. In the small corner of the room, grab the HANDGUN BULLETS. On the desk near the unconscious jailer is a COMBAT KNIFE. Once you have all the items, you may exit the room.[41]

As we can see, the walkthrough is full of things we need to do in order to get out of the cell and to the graveyard, but these things are not worth telling in the previous narration of the plot (in the same way that what the guard says is not worth telling in the walkthrough, which merely states "this will prompt a cut-scene").

When we say that both levels are "active" in our minds at the same time, we do not mean that they are as clearly separated as in our descriptions here. This is an abstraction necessary for analysis, but in fact the act of playing is more than the sum of the two, since actual gameplay is full of doubts, ineffective movement, reloads, and at some point, probably death. This content is also an essential part of playing.

So the act of playing is informed by: our cognitive, often unconscious filling in the gaps, our sense of what we think we have to do within the game,

and the hand-eye coordination that gives us our playing ability. Below is Tosca's description of the first time she played the opening scene of *Code: Veronica*, demonstrating the messy reality of actual gameplay:

> I walk around the cell in darkness, it takes some time to find lighter and learn how to operate it. I watch the cut-scene, have no idea what the guy is talking about, if they have been attacked shouldn't the attackers be on my side? Why does he say I have no chance? Maybe it is a trap, am not sure I should trust him but have no option. He doesn't talk anymore even though I try to click on him and approach him. (I don't see the herb they recommend to take in the walkthrough.) I explore a lot of things with no result (try to open cupboard and get closer to the chest of drawers, but it doesn't work). Since I know the game is about zombies and that the guy belongs to the evil corporation, I am not sure he won't turn into a zombie at any minute and kill me, so I stab him repeatedly to finish him off before leaving the cell. This has required equipping the knife instead of the lighter, something that needs several tries before I find out how to do it.

Actual gameplay is full of trial and error—especially at the beginning of the game when we are not familiar with the interface or the story. My description of the endless things I attempted (moving the body, looking in the cupboards), and my rationale for stabbing the man (the implicit association with movies or other games in which the undead come back to haunt you), are all actions appropriate to the unfolding framework of the story. Though they differ from the sequence of actions recommended by the walkthrough above, they are not wrong; the player is simply using the information that she has at that moment as she tries to fill in the gaps. The player will eventually fill the gaps in the right way and find the right sequence of actions to perform (alone or with help of a walkthrough); or, perhaps, she will quit the game in frustration. And here is a remarkable feature of narrative in video games: it is perpetually unfolding, constantly folding back on itself, full of false starts and restarts, as the player contributes to the story's creation with each action.

Our take on the player experience of narrative resonates with Gordon Calleja's more recent concept of "alterbiography," defined as the "active construction of an ongoing story that develops through interaction with the game world's topography, inhabitants, objects, and game rules and simulated environmental properties."[42] Calleja is also influenced by Iser, and talks about how the scripted narrative inspires and becomes a part of the player's alterbiography, creating narrative involvement.

We hope we have demonstrated that a close reading of a video game is not only possible but also that it yields interesting insights about how a game that makes use of stories is experienced by a player. Of course, one should keep in mind that the analysis presented here is very much determined by the genre to which *Resident Evil Code: Veronica X* belongs, just as any similar analysis will be tied to its subject. Different subgenres will require an

adaptation of these concepts, but it is our belief that they are flexible enough to be extensively applicable. Using the notion of the "literary repertoire" demonstrates the vital importance of a game's cultural (literary, filmic) context, and its gameplay context (knowledge from other games and genres). Similarly, exploring our innately human tendency to "fill in the gaps" of a story shows us that the interpretive process goes hand in hand with the skill-based operation of the game controls, which we can also call interactivity. Drawing from these and other insights of reader response criticism, we are given an intriguing perspective on narrative video games, one in which story and game are not opposed to each other but merge in a unique combination.

A BRIEF HISTORY OF LITERARY THEORY AND VIDEO GAMES

Having examined some of the applications of ideas about narrative to video games, here we will offer a historical look at the theoretical work that explicitly deals with questions of narrative, storytelling, and fiction in relation to video games.

It is always arbitrary to isolate particular time periods, but there is a basic distinction between the works that appeared in the 1990s (or before) and those that appeared from 2000 onward. The 1990s was an exciting time in the nascent field of digital aesthetics: computers had become widespread, the Internet was becoming a major cultural focus, and various forms of digital art, literature, and games were emerging. A whole group of theorists was attracted to the new digital medium, determined to uncover in what way it was special and different from previous media. These texts tend to consider video games as part of more general theories about digital media, while later works deal with video games as a unique object of study.

The transition is best demonstrated by Espen Aarseth's groundbreaking book *Cybertext*, from 1997, still a landmark in the theoretical study of new media and narrativity. He makes a sharp analysis of the changes in textuality (the nature of the text) brought about by the digital era. Aarseth proposes a new model of textuality that can accommodate (among other things) print books, hypertexts, and video games, which he classifies according to an innovative typology. According to Aarseth, video games are "cybertexts," that is, "texts that involve calculation in their production of scriptons,"[43] which means that the participation of the reader/user is not trivial, and the degree of involvement goes beyond the activity of reading. Aarseth's foundation is **narratology** and reader response theories, which he develops to fit the digital medium. His main point is that reading a normal book requires an interpretive effort on the part of the reader, but engaging with a cybertext requires something more, as it is a configurative activity. Cybertexts have existed for a long time (the *I-Ching*, for example, a book that uses printed symbols to decode the tosses of a group of coins, has been around for nearly three millennia[44]), but only with the computer do their properties become widespread and easy to implement.

Narratology: The study of narratives. Within computer game research narratology is often seen as opposed to ludology.

Cybertext examines the main genres of cybertextuality available at the time: hypertext, video games (in the form of adventure games), machine-generated literature, and MUDs. It was an important work in that it opened the textual paradigm to the texts of the digital era, offering an attractive alternative to the dominant hypertext theory of the time (which emphasized the similarities of the new kinds of text with postmodern theories of dispersion).

Other pioneering work in the area was done by Janet Murray, in her *Hamlet on the Holodeck*, also from 1997, which we will consider later in this chapter; also by Marie-Laure Ryan[45] and Mark J. P. Wolf,[46] who attempts to describe video games as an independent medium within a framework of cultural and media history, leaning heavily on film studies, although he also considers the "formal aspects" of the video game (space, time, and narrative).

Both of these first theorists and those who came after have been concerned with a number of issues very much framed by the typical theoretical problems of the digital era. These have kept hypertext theorists and digital aesthetic critics busy since the early 1990s:

Ludology: The study of games, particularly computer games. Ludology is often defined as the study of game structure (or gameplay) as opposed to the study of games as narratives or games as a visual medium.

- **Ludology** versus narratology
- Interaction versus narrative
- The interactive storytelling paradigm: in search of quality
- The fictional status of video games.

We can use these headlines as a way to thematize the theoretical discussions in the field.

Ludology Versus Narratology

From our earlier discussion of literary theory, it seems like looking for stories in video games would be quite straightforward, utilizing the accumulated knowledge of literary theory in order to approach our new aesthetic object. But this approach has not been uncontested. Quite the opposite—the use of literary theory has caused the biggest clash of paradigms in video game studies so far: the "war" between ludology and narratology.

Literature is one the oldest subjects of study in the humanities, and as such its methods have been exported to the study of newer media, such as cinema; first steps in new disciplines are usually inspired by older ones. We could say that the same has happened in the case of video games, as some early approaches to the study of games centered on their representational quality, and thus authors have been able to apply literary and dramatic models to the description and cultural understanding of video games.[47]

However, some of these attempts have been rejected by a group of researchers, identified (by themselves and others) as "ludologists."[48] Their aim is to consider video games as games and not as narratives or anything else. The term "ludology" was introduced into video game studies by game

theorist Gonzalo Frasca. He suggested that ludology be thought of simply as the "discipline that studies game and play activities."[49] In that broad sense there is no inherent opposition to literary approaches, and indeed "ludology" *is* sometimes used to merely refer to "the study of games" (using any approach). Expanding on this stance, however, Frasca has since specified that a ludological stance implies that "games cannot be understood through theories derived from narrative."[50]

Jesper Juul's article "Computer Games Telling Stories?" offers a balanced introduction to the debate. He argues that part of the problem arises from the indiscriminate use of the word "narrative." If everything is a narrative, it is of course not very useful to say that video games can also be described as such. Juul also gives three reasons for why it is easy to confuse narratives with video games:

1. The player can tell stories of a game session.
2. Many video games contain narrative elements, and in many cases the player may play to see a cut-scene or realize a narrative sequence.
3. Video games and narratives share some structural traits.[51]

Juul argues that we should resist the temptation to confuse the two forms, because it is impossible to translate video games into stories and vice versa.[52] He also discusses how understanding narrative simply as a retelling of events can be misleading and how the experience of playing a game is very different from that of reading a story.

Ludologists have found a wide variety of reasons why subsuming video games under the headline of narrative is a bad idea; the most damning is that the approach can lead us to overlook the "intrinsic properties" of video games. The Finnish theorist Markku Eskelinen has overtaken Frasca's less confrontational stance and is in the vanguard of ludology's most radical anti-narrative critics. As he forcefully puts it:

> The old and new game components, their dynamic combination and distribution, the registers, the necessary manipulation of temporal, causal, spatial and functional relations and properties not to mention the rules and the goals and the lack of audience should suffice to set games and the gaming situation apart from narrative and drama, and to annihilate for good the discussion of computer games as stories, narratives or cinema.[53]

But the extremity of such a stance has further stoked the debate. The idea that games are first and foremost games—an argument bordering on the tautological—can be taken to mean that the formal properties of video games are more important, more *intrinsic*, than the stories in the games (or the graphics, or the social activity that games promote). The debate has raged on for the last several years, with ludologists attacking the colonization of the new field of game studies by alien disciplines such as literature. Controversy is sometimes good, and from a pragmatic perspective we can

say that this discussion has helped bring the study of video games into the spotlight.[54]

The debate has continued among game scholars, although lately the ludologists have been identified as "essentialists," and narrative is no longer the enemy. The new struggle is between those who prefer to concentrate on the formalistic properties of games (the ludologists) and those who want to consider them as situated practices.[55]

To return to the focus of this chapter, literary theorists have criticized radical ludology for viewing stories as "just uninteresting ornaments or gift wrappings to games, and laying any emphasis on studying these kinds of marketing tools is just a waste of time and energy."[56] Scholars like Marie-Laure Ryan, and later Julian Kücklich,[57] see ludology as a threat to the fruitful application of narrative concepts to the study of video games.

Computer games function as what Kendall Walton would call a "prop in a game of make-believe." It may not be the raison d'être of computer games, but it plays such an important role as a stimulant for the imagination that many recent computer games use lengthy film clips, which interrupt the game, to immerse the player in the game world.[58]

From this it follows that the two sides seem to be talking about different things when they say "narrative," as ludologists see it as a fixed sequence of events forced on the player, while literary theorists see it as a loose fictional frame.

Paradoxically, certain ludologists have come to agree with the importance of stories. Jesper Juul's *Half-Real: Video Games between Real Rules and Fictional Worlds* represents a move away from his earlier radical position, as he argues that the essence of video games as a cultural product is the unique way in which they combine "the rules that the player interacts with in real time with a fictional, imagined world."[59]

Beyond the squabbles, we should recognize the ludology-narratology wars as a symptom of the struggle to define the new discipline of game studies, beyond the dominant paradigms—the hypertextual[60] and the cinematic[61]—of the beginnings of the study of digital textuality.

Interaction Versus Narrative

A game has to be interactive. This is the greatest strength of video games, but it is also a challenge. If the point of video games is that players do things (gameplay), the fact that in some video games they are forced to "sit and watch" (for example, cut-scenes) is seen by some designers and others as an unwelcome invasion. And if designers are able truly to integrate stories into the game, how can narratives ever be fully engaging if they always have to offer the player the possibility of winning? And how can artificial intelligences ever become interesting fictional characters? All these questions are derived from the ongoing debate of interaction versus narrative, a problem inherited from earlier discussions about the status of digital literature, mostly from the field of hypertext theory.[62]

One solution to this problem is to separate interaction from narration and situate them in different planes, as Janet Murray does when she conceives of video games as symbolic dramas:

> In games, therefore, we have *a chance to* enact our most basic relationship to the world—our desire to prevail over adversity, to survive our inevitable defeats, to shape our environment, to master complexity, and to make our lives fit together like the pieces of a jigsaw puzzle. Each move in a game is *like* a plot event in one of these simple but compelling stories. *Like* the religious ceremonies of passage by which we mark birth, coming of age, marriage and death, games are ritual actions allowing us to *symbolically* enact the patterns that give meaning to our lives.[63]

The trouble with this answer is of course the *like*, which puts video games and stories at different abstract levels. This might explain how narratively we make sense of the conflicts posed by video games but doesn't tell us much about how stories actually work *within* them. We could also say that it seems much easier to create symbolic dramas than real interactive stories (at plot level), since anything can be read as a symbolic drama. Even *Tetris*, after all, has "clear dramatic content."[64]

An alternative is Atkins's analysis of video games that effectively integrate narrative and gameplay. *More than a Game* is dedicated to four video games that he sees "as having a central narrative impetus, that develop story over time, rather than simply repeat with minimal difference in a move from level to level of increasing excess."[65] The chosen games are *Tomb Raider, Half-Life, Close Combat*, and *SimCity*—which seem like strange choices since there are no adventure or role-playing video games among them, the two subgenres with a stronger narrative component.[66] Each of the four games represents a good example of an effective strategy for integrating narrative and gameplay: *Tomb Raider* is a model of open-endedness and the use of quests; *Half-Life* employs innovative themes and perspective; *Close Combat* takes the concept of realism to new heights; and *SimCity*'s simulation rules can be read narratively, which means that game rules can have ideological implications. The book concludes with a discussion on the status of video games as a fictional form. It is a very compelling approach, but no links to previous theory are made, which undermines the author's credibility when he tackles highly contested issues in the field of digital culture—such as interactivity or immersion—as if for the first time.

The narrative versus interactivity debate also crops up in most video game design textbooks, although the authors might not identify it as such.[67] For instance, game designer Richard Rouse argues that all out-of-game storytelling should be avoided, as it goes against the interactive experience of gaming. Rouse uses examples from bad cut-scenes (for example, from the game *Tekken*) to argue that very often they are not integrated into the game.[68] In-game storytelling is the preferable option, which Rouse describes as "showing," compared to the "telling" which occurs with out-of-game

narrative. In-game storytelling is achieved by, for example, creating a believable world that gives a context to the plot, and by nonplayer characters with personalities. *Deus Ex*, with its successful in-game storytelling, is a good example of how to achieve both narrative and interaction; the gameplay offers the choice of acting in a violent or more pacifist way, so that players can create different stories as they go. "For in-game storytelling, players get to experience the story themselves instead of being told it secondhand"; it also maintains the player's immersion in the game world.[69]

Rouse is not alone in proposing a way to use "storytelling" to thematize the rules in games. Designers and theorists alike argue that fiction might actually have a very important function in video games.[70] This connection is explored by Jesper Juul by looking at examples of "clashes" between rules and fiction:

> In the game of *Grand Theft Auto 3*, a blue arrow is placed over whatever object or target the player has to reach in a mission. Even though the game has been lauded for its elaborate world and detailed environments, the blue arrow clearly points to the *rules* of the game. This is at odds with a certain mode of thought that claims that a good game should never remind the player that he/she is playing a game. I claim that the player is always aware that the game is a game, and that this is part of the reason why we can accept incoherence in game worlds.[71]

Juul argues that the fictional world is also needed in video games to enable players to infer the rules (to let them know which repertoire applies)—which in "good" games are anyway hidden from immediate perception.[72] This compromise is a step forward from early ludological positions, and certainly demonstrates a more effective way of reconciling literary theory and video games.

Perhaps further reconciliation is possible through a reexamination of our terminology. As we have already suggested, the idea that narrative and interactivity oppose each other springs from a definition of narrative as a linear unfolding of events; while this is true in many instances, it is not very interesting, as Jonas Heide Smith remarks.[73] There are other forms of narrative than the traditional sequence, and especially interesting possibilities have recently bloomed, with the growth of multiplayer online games. We might even speculate that this argument will soon be a thing of the past, as we move past the dominant influence of single-player games. As we move forward, we might want to look at how narratives are produced in multiplayer environments, for example, in which the parameters of the story are not all fixed by a designer and where new kinds of experiences arise due to the unpredictability of the players' behavior.[74]

In the same spirit, Henry Jenkins argues that games tell stories in their own unique way, and that their narratives are not equivalent to a simplistic, linear idea of a story of the type found in films or novels.[75] Jenkins proposes, instead, that game stories are "spatial" and "environmental." Spatial

stories are related to those works of literature that tell a hero's odyssey, or travel narratives; in the case of computer games the traveling would be done not only by the characters but also by the player. Environmental stories are related to the genres of literature that focus on the depiction of worlds, such as fantasy or science fiction; in computer games, it is the player that does the exploring. In order to construct a world that tells a story, game designers act like creators of amusement park rides; they are not storytellers but architects of narrative. Accordingly, Jenkins describes four ways in which narrative can appear in games:

- *Evoked narratives*: When video games reproduce a world that is known to players through other works of fiction, so that the games are part of an encompassing system of meanings. For example, the game *Blade Runner* reproduces the world of the film and the novel, so that players will be able to "live out" the stories that exist in their imagination by visiting that world.
- *Enacted narratives*: Where, as Jenkins puts it, "the story itself may be structured around the character's movement through space and the features of the environment may retard or accelerate that plot trajectory."[76] This category encompasses the spatial and environmental stories mentioned above. These are typically not appreciated by literary scholars, as they privilege spatial exploration over plot development. This form is very well suited to video games, as a designer can create the world that the player will have to traverse and conquer.
- *Embedded narratives*: When "the gamespace becomes a memory palace whose contents must be deciphered as the player tries to reconstruct the plot."[77] The classical embedded narrative would be a detective story, such as those found in adventure games, where players encounter a world of clues that have to be deciphered in order to find out about the story that has already happened. A good example of this is *Myst*.
- *Emergent narratives*: When "gamespaces are designed to be rich with narrative potential, enabling the story-constructing activity of players." We have seen Juul's description of these narratives earlier, in the section on "Mechanics"; both Juul and Heide Smith have called this "object-oriented narration," in which the gamespace is filled with objects with their own behaviors, so that the player interaction creates unique events that feel significant. Henry Jenkins gives the example of *The Sims*, which allows players to create their own stories.[78]

These four kinds of narratives fail to address many of the elements discussed in the first part of this chapter, such as the sequential storytelling we find in quests, or other literary ingredients such as characterization. However, they may help move us beyond the frozen narrative versus interactivity discussion. Because these categories highlight other qualities of video game stories, such as their spatial component, and provide a way to introduce contextual elements (using the evoked narratives category, for example).

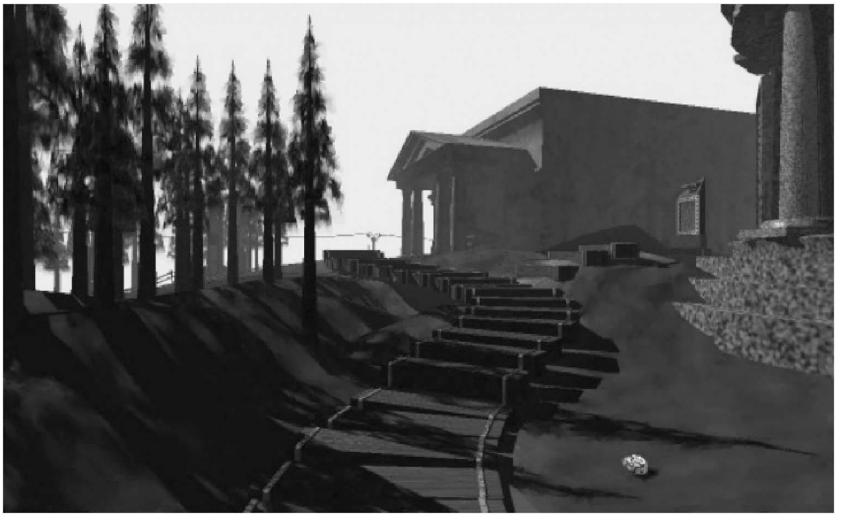


Figure 7.8 Myst (1994), with its special embedded narrative, attracted a new audience to video games

They offer a different and fruitful use of literary theory in relation to video games.

The Interactive Storytelling Paradigm: In Search of Quality

During the 1990s, the idea of the "interactive narrative" gained a certain prominence. It refers not to video games, nor hypertext (or other forms of digital literature) but rather to a sort of role-playing in which the computer can anticipate the player's reactions and give an illusion of total interactivity. It is considered an ideal form of digital storytelling, deeply engaging and immersive. This paradigm approaches narrative from an Aristotelian and structuralist perspective, and its proponents are typically interested in the problems inherent in automatic generation of stories, or how to improve artificial intelligence. The field of interactive narrative has attracted scholars that work with design and programming of games, such as Craig A. Lindley or Michael Mateas,[79] as well as theorists such as Marie-Laure Ryan.[80]

The most influential work in this area is *Hamlet on the Holodeck*, by Janet Murray, an enthusiastic defense of the computer as the new storytelling medium, because of the way its four defining properties facilitate digital narration:

> Digital environments are *procedural, participatory, spatial* and *encyclopedic*. The first two properties make up most of what we mean by the vaguely used word interactive; the remaining two properties help to make digital creations seem as explorable and extensive as the actual world, making up much of what we mean when we say that cyberspace is *immersive*.[81]

The "procedural" property refers to the computer's ability to execute rules in succession and thus generate behaviors. For Murray, the computer character Eliza, created by Joseph Weizenbaum in 1966, is a good example of how the procedural property can produce the illusion of a "living person." Eliza was able to converse with humans as if she was a psychologist that always turned your statements back at you. For example, if you typed: "I am worried," the program would respond: "What are you worried about?" Inspired by this, authors like Ian Bogost and Noah Wardrip-Fruin have developed theories about the specific expressive power of video games (and other digital cultural forms) that are specifically tied to the procedural property (see Chapter 6).

The second, "participatory" property refers to the fact that we can influence the production of behavior when interacting with a computer; an example is the textual adventure game *Zork*, in which the program responds to the textual input of the player and produces lines of text that describe her progress in the game, thus creating a story. Computers are also "spatial," that is, able to represent navigable space.

Finally, the "encyclopedic" property points to the storage capacity of computers, whose memories can hold enormous amounts of information and allow us to interact with information-rich environments, from fan websites to MUDs to enormous games like *SimCity* and *Civilization*.

Without going into a detailed discussion of the four properties, it is not hard to see that video games turn out perfectly to embody each of them, and are thus, it could be argued, the perfect vehicle for digital storytelling.

Murray is inspired by the "holodeck," the entertainment system of the *Star Trek* universe. The holodeck is an immersive entertainment environment, a virtual reality machine that makes stories come to life and lets "readers"—in this case the ship's crew—participate as actors in completely realistic virtual environments that make use of all five senses. An example Murray gives is that of the female starship captain in the series *Star Trek: Voyager* who is entertained with a story inspired by *Jane Eyre*, in which she plays a nineteenth-century governess and nearly falls in love—in "real" life—with the virtual hero.[82]

Following the promise of the holodeck, Murray's dream is that of a "cyberbard"—a Homer or Shakespeare of the future, able to bend the computer to more artistic uses than those we know today. Murray's book is inspiring, filled with rich personal anecdotes about new media experiences, even though it also suffers from excessive utopianism. The book overstates the interest that interactive narrative might hold for average audiences, and projects the idea of the romantic author into the future. As for computer games, even though Murray identifies them as the greatest commercial success and creative effort in digital storytelling, she thinks they are not yet mature as a medium: "The narrative content of these games is thin, and is often imported from other media or supplied by sketchy and stereotypical characters."[83]

Janet Murray was influenced by an earlier book by Brenda Laurel, *Computers as Theatre*, from 1991, about interface design and drama, although

it also deals with video games and offers a dramatic model for computer-human interaction. Laurel's opinion of video games, however, regardless of their technological advancements, is also not very positive: "I find most video games to be boring, frustrating and 'obstructionist' in the sense that they require players to solve puzzles primarily for the purpose of extending the duration of game play."[84]

The dubious opinions these scholars hold of video games are in part due to their judging them by the quality standards of other media, such as movies or print novels. They suggest that the simplicity of game narratives can be overcome in different ways, involving everything from complex artificial intelligence or more detailed fictional worlds. Mark J. P. Wolf, another author worried about the quality of video games, seems to equate the emotional complexity of a player's experience with the level of detail in the games' visual imagery. He writes, for example, that *Myst* is a turning point in the development of the video game's diegetic world, because of the light effects that create a rich world and that inspire contemplative moods in the player.[85]

Murray offers some guidelines for creating engaging, interactive storytelling. To begin with, formulaic storytelling could make use of the same plots all stories are made of, combined in an interesting way that possibly involves powerful artificial intelligence.[86] Murray suggests that procedural storytelling could be similar to ancient oral storytelling (the bardic system), although how this inspiration can be used to combine plots is not entirely clear. She also stresses the importance of using well-known genres (like mystery or science fiction) in order to exploit knowledge that the user already has—in a way similar to the "repertoire" concept of reader response theory.

Vladimir Propp's "morphology of the folktale" might be another tool to help create digital stories. His system, created in 1928, describes how Russian folktales can be analyzed by isolating small narrative units that are combined in different ways but essentially repeat themselves in many tales (for example, the hero leaves home, or the hero gets a present). Murray argues that interactive narrative would benefit from reproducing his formulaic system of story elements, symbols, and character functions. According to Murray, Propp's algorithm is much more complex than "most electronic computer games currently on the market," which are limited to two or three themes such as "fight bad guy," "solve puzzle," and "die." Without considering the economic reasons behind this simplicity, she sees it as a problem that video games "do not allow substitution of thematic plot elements" that would require much more ambitious programming.[87]

A productive take on these problems is exemplified by the work of scholar and designer Michael Mateas, who together with Andrew Stern has created the most successful interactive drama to date, *Façade*. This piece combines both a credible and engaging plot with believable characters, while allowing real interaction on the part of the player, who communicates with the program using natural language.

Mateas and Stern explicitly wanted to create an Aristotelian plot arc.[88] Their AI system tries to respond to the input of players and change dramatic



Figure 7.9 Façade (2005) attempts to use natural language interaction to create a deeper interactive narrative experience

values so that the interaction actually has an effect on the development of the story. What lies behind the relatively simple interface is a carefully planned programming of story units, which they call "beats." "Roughly, a beat consists of an action/reaction pair between characters. Beats are sequenced to make scenes, scenes to make acts, acts to make stories. The AI plot system contains a library of beats appropriate for our story."[89] *Façade* is both a simulation and a plotted narrative. At any moment, the player experiences the freedom to interact in natural language and to send the action one way or the other, but a "drama manager"

> continuously monitors the simulation and proactively adds and retracts procedures (behaviors) and discourse contexts by which Grace and Trip operate. That is, the rules of the simulation are regularly being updated in an attempt to give the player a well-formed overall experience with unity, efficiency and pacing.[90]

And it works. Even though there are occasions on which the system fails to recognize player input, most interactions elicit an appropriate answer, and the story can be played a few times with different outcomes.

The paradigm of the interactive narrative does have a lot to offer our study of video games. In the relentless search for quality advocated by Murray, Laurel, and others—in their insistence on the ways that narrative can be adapted to new technologies—we have to admire their enthusiasm about the possibilities of the new medium and their attempt to describe what it is about interactivity that people desire so much.

The Fictional Status of Video Games

From the foregoing discussion, it seems clear that fiction in video games does not work according to the same parameters as in representational media; it should also be clear that we cannot apply literary theories wholesale to the study or design of video games without considering their specific properties. A variety of authors have begun to move beyond the quagmire of narrative, to focus instead on what kinds of fictions video games are, and how we could delimit their fictional properties.

A foundational theoretical premise for these scholars is to decide what the game world is. Is the game world a prop for the player's imagination (in Walton's sense), or is the world a virtual object or stage for the game to take place on? In Marie-Laure Ryan's words:

> In an abstract sense, of course, most if not all games create a "game-world," or self-enclosed playing space, and the passion that the player brings to the game may be regarded as immersion in this game-world. But I would like to draw a distinction between "world" as a set of rules and tokens, and "world" as imaginary space, furnished with individuated objects. The pieces of a chess game may be labeled king, queen, bishop, or knight, but chess players do not relate to them as fictional persons, nor do they imagine a royal court, a castle, an army, and a war between rival kingdoms.[91]

One of the consequences of such a distinction is that the world we imagine in our heads is not equivalent to the visual appearance of the virtual world of the game. Thus, we should consider the whole world experience and not just what has been called, by authors such as Wolf, the "diegetic representation" of a world, which is just the "world on screen."[92] For Wolf, this is different from "extradiegetic narration," that is, how video games point to other diegetic worlds (television shows, movies, novels, and the like). Wolf's examples of extradiegetic narration are of intro movies, cut-scenes, intro screens, and game manuals. However, the meta level of genre references not only points to stories in other media, it also calls up the player's game repertoire[93]—as discussed in relation to reader response theory—so that the player can also infer how the game world will work. In other words, the world allows us to make inferences about the rules.

Salen and Zimmerman's concept of the "narrative descriptor" is a good example of elements beyond the game. Narrative descriptors are "representations, which means that they are depictions of one or more aspects of the game world";[94] these can be graphics on the side of the arcade cabinet, manuals, soundtracks, cut-scenes, and so on. Let us, for example, look at their description of the action in *Asteroids*, from which players will be able to infer much about interaction and gameplay:

> Shoot the asteroids while avoiding collisions with them. Occasionally a flying saucer will appear and attempt to shoot you down with guided missiles. Destroy it or the missiles for more points.[95]

These few sentences remind us not only of a genre that is related to other media—science fiction—but also of typical video game conventions—in this case the reward system. Video game conventions, for example, having several "lives," are very important for the gaming experience, even if they wouldn't make sense in a traditional narration. Imagine an account of *Romeo and Juliet* where Romeo was allowed to start all over again after discovering his lover dead, as many times as it took for them to be together and live happily ever after. We would perceive this not only as incoherent but also as totally destructive of the play's dramatic effect. Jesper Juul argues that the incoherence and instability of video game fictions does not mean that these fictions are of lesser quality than those found in other media; rather, they just need to be considered from their own perspective:

> The worlds that video games project are often ontologically unstable, but the rules of video games are very ontologically stable. While we may not be able to explain *why* Mario reappears in *Donkey Kong*, we always know for certain how many lives we have left. That the majority of fictional game worlds are incoherent does not mean that video games are dysfunctional providers of fiction, but rather that they project fictional worlds in their own flickering, provisional, and optional way.[96]

Espen Aarseth has also dealt with the question of fiction in video games, and comes to the conclusion that computer games are not fictions in the same way as, say, books or films are, even though they do contain fictional elements. Some of the fictional elements in computer games are just representations (for example, a door that cannot be opened), but others can be operated in the same way as objects in the real world (some doors can be opened, closed, and locked). These would have a middle status, a combination of representation and simulation. Aarseth talks about "virtuality," and concludes that "instead of the common notion that game worlds are fictional, we should start to see them as composites where the fictional element is but one of the many types of world-building ingredients."[97]

In a remarkable conjunction of ideas, other authors from Henry Jenkins to Julian Kücklich have also proposed literary approaches to the study of video games based on their capacity to evoke fictions, and not on their narrative qualities,[98] which takes us out of the cul-de-sac of the narratology-ludology debate. The fictional worlds approach has also opened the door to studying video games as a part of wider ecologies of fiction, such as transmedia worlds, where the specific strengths of video games for storytelling purposes are considered in relation to other media products such as books, films, or television series.[99]

TRANSMEDIA GAMES

Transmedia studies have become a field in its own right[100] and grown very much in scope since Marsha Kinder identified video games as a crucial part of what she called "commercial transmedia supersystem,"[101] which we today

might rather call transmedia franchises. Kinder proposed that intertextuality could emerge from products such as toys and of course games. Also Henry Jenkins, who coined the expression "transmedia storytelling," considered games as one more brick in complex entertainment systems, where

> In the ideal form of transmedia storytelling, each medium does what it does best-so that a story might be introduced in a film, expanded through television, novels, and comics, and its world might be explored and experienced through game play.[102]

This somehow subordinate position of video games (at least in story-related terms) has been the rule whenever video games have been part of complex transmedia systems, also today, they are "characterized by an adaptation aesthetic, a process through which the images and characters from the film become playable and/or navigable."[103] Transmedial studies as a field spans over several disciplines, ranging from media studies to communication to the humanities, so video games are interesting objects of study from different angles (as part of franchises, as carrying specific affordances, as fulfilling narrative roles). The narrative-centric transmedia research is best represented by the work of Marie-Laure Ryan[104] and Jan-Noel Thon,[105] who both have argued in favor of separating the concept of transmediality from an attachment to a particular medium and instead work toward an overarching theory of transmedia narratives that would also include video games:

> I am thinking here of principles such as extending the timeline, creating prequels and sequels, telling the story of secondary characters, extending the geography of the storyworld, telling the story from a different point of view and leaving some unresolved issues that can be answered in another narrative. These principles are timeless and independent of medium.[106]

FINAL REMARKS

Perhaps these developments mean that the field is moving toward more stimulating discussions that also take into account more contemporary literary theory. However, postmodern literary theory has not been a part of the debate so far, and perhaps it never will be. Stories in computer games are still in their infancy, and the heavily loaded postmodernist theories about deconstruction or other dispersions of the text wouldn't be of much help at this stage.

So let us return to the question we posed at the beginning of the chapter: are video games stories? Yes, we would answer. Many video games are stories, as well as games. Some games contain more narrative than others, but even the most abstract usually include the sketched elements of a fictional world. And we need to take into account the narrative elements of a video game if we are to fully understand the medium and how player interaction is shaped.



DISCUSSION QUESTIONS

- Think about a game you have played where you liked the story. How is it different from other kinds of stories, such as those in books or movies? What do you like and dislike in video game stories?
- Think about a memorable video game character (including player characters) and discuss why you remember him or her. How are characters in games different from characters in other media?
- What makes a good quest in a game? Why? Use examples to illustrate your answer.
- Summarize a good story-based game you have played. What do you leave out when you summarize it? Why?
- Are interactivity and narrative totally opposed? In other words, can a game designer give options to her players while at the same time making sure they experience a good story? Discuss, using theory from the chapter as well as examples from your own experience.



RECOMMENDED GAMES

Resident Evil Code: Veronica X—A great example of the narrative strength that can come to play in computer games.

BioShock series—A great example of how narrative can work in games, and where the moral nature of your choices adds an extra dimension to the game experience. *BioShock* shows how the story and the game can really work together, where too often the narrative almost gets in the way of the game.

Baldur's Gate 2—A classic example of how narrative is used in games to enrich the classic game mechanics. Very well executed with a few interesting twists where the narrative sometimes breaks the artificial boundary between the game and the story—allowing the story to overtake the game experience at key points in the story.

Façade—An early example of interactive fiction heavily relying on artificial intelligence to construct a unique experience every time you play the game. Often seen as one of the first successful examples of a truly interactive narrative able to respond to player actions.

Silent Hill 2—An intriguing example of how well narrative can work to infuse the game experience, where your actions in the game influence how the story develops. The twist of the narrative undermining your protagonist is as disturbing as it is effective in creating a unique experience.

FURTHER READINGS

Aarseth, E. (2012, May). A narrative theory of games. In *Proceedings of the International Conference on the Foundations of Digital Games* (pp. 129–133). ACM.

(Estimated reading time: 22 minutes)

Fernández-Vara, C. (2011). Game spaces speak volumes: Indexical storytelling. *Proceedings of DiGRA 2011 Conference: Think Design Play*.

(Estimated reading time: 28 minutes)

Kücklich, J. (2001, September). Literary theory and computer games. In *Proceedings of the First Conference on Computational Semiotics for Games and New Media (COSIGN)*.

(Estimated reading time: 35 minutes)

NOTES

1. From the game package of Westwood's *Blade Runner*. Just like the film, the game is set in Los Angeles in 2019, and the player has the role of a rookie blade runner named Ray McCoy, who has to deal with several dangerous replicants.
2. www.lucasarts.com/products/starwarsracer/default.htm.
3. On the similarity of *The Sims* and doll's house play, see Flanagan, 2003; Krotoski, 2004, p. 12.
4. From promotional material for the game, as quoted at, for example, www.metacritic.com/game/gamecube/the-sims.
5. The discussion here builds on Rimmon Kenan, 2002.
6. As Rimmon Kenan herself notes, these distinctions have been given different names by different literary theorists; for example, Genette's *histoire, récit*, and *narration*. This is not the place to discuss them and compare them, as we are only interested in definitions that will help us understand computer games.
7. In her PhD thesis, *Towards a Poetics of Virtual Worlds: Multiuser Textuality and the Emergence of Story* (Klastrup, 2003, p. 19).
8. Aarseth, 2000.
9. Aarseth, 2000.
10. Calleja, 2011, p. 92.
11. Rouse, 2001, p. 223.
12. Klevjer, 2002; Wolf, 2001.
13. Klevjer, 2002, p. 198. This definition follows Gérard Genette, for whom "diegesis" would mean the narrative of a work of fiction, that is, the story, so that everything that did not belong to the main story would be "extradiegetic." Later, in film theory, diegesis has been understood to include the elements that belong to the main story (for example, the background of the characters that has led to their situation).
14. Klevjer, 2002, p. 200.
15. *Collins English Dictionary*. Italics added.
16. Wolf, 2001, p. 98. Unlike characters in other media, who are deeper and capable of development, characters in video games follow a goal, which can be "score oriented, conflict oriented, task oriented or some combination of these." However, Wolf's argument suffers from a confusion between the personality of the fictitious characters (in many cases absent) and the actions players

have to take when controlling them. There is no reason why a playing character cannot be a narrator, for example.

17. Vladimir Propp published his *Morphology of the Folk Tale* in 1928, offering among other things a typology of characters according to their function in these narratives, such as: the hero, the villain, the donor, the dispatcher, the false hero, the helper, the princess, and the father (of the princess); Propp, 1969.
18. Gard, 2000.
19. MacCallum-Stewart and Parsler, 2008.
20. Tosca, 2003a.
21. Rouse, 2001, p. 235. However, some theorists disagree. Ken Perlin, for example, maintains that we cannot identify with game characters unless they have a strong personality, since if they are empty and only "filled" through our agency, identification is impossible; Perlin, 2004.
22. See, for instance, Bringsjord, 2001.
23. Laurel, 1993, p. 145.
24. Meehan, 1976.
25. Lebowitz, 1984.
26. Lebowitz, 1984, p. 146.
27. Smith, 2000a.
28. Juul, 2001.
29. Smith, 2000b.
30. Tronstad, 2001; Tosca, 2003b; Aarseth, 2003; Howard, 2008. The analysis here is based on Tosca, 2003b.
31. Walker, 2007.
32. Howard, 2008.
33. Krzywinska, 2009.
34. Ryan, ML (2001b).
35. Released in 2001 for the PlayStation as an updated version of the original 2000 Dreamcast game.
36. Iser, 1979, p. 69.
37. "Actualize" here means to understand the cues provided by the text and call up the relevant repertoire—in other words, make the appropriate projections.
38. Iser, 1979, p. 169.
39. In Marie Laure Ryan's sense; Ryan, 1991.
40. From a 135-page document entitled "A Detailed Plot Analysis of the *Resident Evil* Videogame Series," by Dan Birlew and Thomas Wilde. Downloaded from http://db.gamefaqs.com/console/psx/file/resident_evil_plot.txt (June 30, 2005).
41. This walkthrough by CVXFREAK was downloaded from http://db.gamefaqs.com/console/dreamcast/file/resident_evil_code_veronica_complete.txt (June 30, 2005).
42. Calleja, 2011, p. 127.
43. Aarseth, 1997, p. 75. "Scriptons" are the text units the reader sees on-screen after having manipulated the text. They are different from "textons," which are all the units that make up the text but which are not necessarily visible to the reader, for example, the programming language.
44. The *I-Ching* is a multi-authored text whose current written version probably dates from around 800 years before Christ. It consists of 64 symbols, each of which contains a main text and six smaller texts. Readers have to toss three coins in order to combine these fragments, and the random result (1 of 4,096 possibilities) is used in predictions and future-telling; Aarseth, 1997, p. 75.
45. Ryan, 1991, 2001b.
46. Wolf, 2001.
47. Laurel, 1993; Murray, 1997; Ryan, 1991, 2001b; Aarseth, 1997.
48. See Frasca, 2003b, for a history of the concept. *Ludus* means "game" in Latin, therefore "ludology" would be the study of games. The (mostly Scandinavian) researchers associated with this term are Markku Eskelinen, Gonzalo Frasca, Aki Jaarvinen, Jesper Juul, and Espen Aarseth.
49. Frasca, 1999.
50. Frasca, 2001b.
51. Juul, 2001.
52. Although he doesn't mention adaptation of fictional universes, which is certainly possible, as in the case of the *Blade Runner* game, even though the same sequence of events is not maintained.

53. Eskelinen, 2001.
54. A good summary of the history of the controversy can be found in Frasca, 2003b.
55. See, for example, Taylor, 2006; Malaby, 2007; Consalvo, 2009.
56. Eskelinen, 2001.
57. Kücklich, 2002, 2003a.
58. Ryan, 2001a.
59. Juul, 2005, p. 145.
60. Landow, 1992.
61. Manovich, 2001.
62. Authors like Landow, 1992 or Douglas, 1992 discuss the break with traditional narrative brought about by hypertext narrative.
63. Murray, 1997, p. 143. Italics added.
64. Murray, 1997, p. 144.
65. Atkins, 2003, p. 20.
66. There might be a historical explanation, though, since the classic genre of adventure computer games is no longer so popular, but its characteristics have been incorporated into other subgenres, such as action-adventure or survival horror.
67. Salen and Zimmerman call this opposition "crafted story" versus "emergent experience"; see Salen and Zimmerman, 2004, pp. 383–384.
68. Rouse, 2001, p. 223.
69. Rouse, 2001, p. 227.
70. For example, Ryan, 2001a; Wolf, 2001, pp. 109–110; or Laurel, 1993, p. 106, who talks about goals as framework for action.
71. Juul, 2003a, p. 146.
72. Juul, 2003a, p. 163.
73. Smith, 2000b.
74. Smith, 2000b. This is a rather underexplored area in computer game studies, with the exception of the work of Anders Drachen (later published under the name Anders Tychsen); Tychsen, Hitchens, Brolund, McIlwain, and Kavakli, 2008, Tychsen, Hitchens, and Brolund, 2008; Drachen and Smith, 2008.
75. Jenkins, 2003a.
76. Jenkins, 2003a, p. 129.
77. Jenkins, 2003a, p. 129.
78. Jenkins, 2003a.
79. Examples of their work include Lindley, 2002; Mateas, 2001.
80. In her books *Possible Worlds, Artificial Intelligence, and Narrative Theory* and *Narrative as Virtual Reality*; Ryan, 1991, 2001b. Ryan's work connects literary theories of possible worlds to the realm of the digital, and engages with literary definitions of virtual reality, simulation, and immersion in relationship to narrative.
81. Murray, 1997, p. 71. Italics added.
82. In the episode "Persistence of Vision"; see Murray, 1997, ch. 1.
83. Murray, 1997, p. 51.
84. Laurel, 1993, p. 167.
85. Wolf, 2001.
86. Murray, 1997, p. 186.
87. Murray, 1997, p. 197.
88. Mateas, 2001, p. 151.
89. Mateas, 2001, pp. 151–152.
90. Mateas and Stern, 2003, pp. 5–6.
91. Ryan, 2001b, p. 307. Jesper Juul also makes a distinction between "the worlds we find in computer games and how computer games cue the players into imagining worlds"; Juul, 2003a, p. 109.
92. Wolf, 2001, p. 94.
93. Tosca, 2003c.
94. Salen and Zimmerman, 2004, p. 399.
95. Salen and Zimmerman, 2004, p. 400.
96. Juul, 2003a, p. 169.
97. Aarseth, 2005, p. 4.
98. Jenkins, 2003; Kücklich, 2003b.
99. Jenkins, 2003; Klastrup and Tosca, 2004, 2011.
100. Freeman and Rampazzo Gambaratto, 2018.
101. Kinder, 1991, p. 38.
102. Jenkins, 2003.
103. Kennedy, 2018, p. 72.
104. Ryan, 2014, 2016.
105. Thon, 2016.
106. Ryan, 2016, p. 7.